KEEGAN DEMPSEY, PE

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WORK EXPERIENCE

NON ALIGNED CLIMATE ACTION PROJECT (NACAP LLC) - New York, NY, USA (November 2023 - Present)

Founder – Independent Research and Advisory

- Advised on technoeconomic model development for two early stage Australian hydrogen startups.
- Designed a decision framework for screening community scale waste to value pathways, integrating feedstock availability, process performance, policy constraints, socioeconomic drivers and local demand into an optimization ready model.
- Built a geospatial data pipeline converting public facility, wastewater, emissions, permitting and transport corridor records into a machine readable network of candidate sites, links and capacity bounds, paired with an interactive viewer for auditing inferred connections before optimization.
- Formulated an optimization tool with technoeconomic analysis at site level, made tractable by a route generation and pruning method that builds candidate pathways from the inferred infrastructure graph rather than a generic superstructure.
- Trained equipment level surrogate models to predict gasification and related conversion yields across heterogeneous feedstocks, reducing reliance on computationally intensive process simulations.

HYSATA - Port Kembla, NSW, AUS (February 2023 - August 2023)

Balance of Plant (BOP) & Systems Engineer / Technoeconomic Modeling Lead

- Modeled load following system models from first principles and experimental electrolyzer performance data to optimize BOP configuration for diverse renewable supply profiles.
- Devised BOP process simulation in Aspen HYSYS integrating dynamic electrolyzer degradation curves to inform equipment sizing, layout and capital cost estimation.
- Ideated, built and managed an integrated technoeconomic model, coupling scenario analysis and performance metrics to quantify the impact of technical and commercial decisions on levelized hydrogen cost.
- Proposed stack level design innovations to consolidate BOP functionalities, achieving cost & performance improvements.
- Aligned technical / commercial team workflows, bridging data and scope gaps to improve cross functional project execution.

ARUP - Melbourne, VIC, AUS & New York, NY, USA (September 2019 - January 2023)

Chemical Engineer – Energy, Transaction Advisory & Sustainability Groups

- Formulated technoeconomic modeling methods, electrolyzer performance models and plant process simulations for incentivizing, sizing and costing of green hydrogen projects worldwide.
- Advised governments and companies on hydrogen and synthetic fuel deployment strategies.
- Substantiated technical due diligence projects, verifying technical aspects of projects and technologies.
- Conducted and oversaw research to develop educational tools for internal and external stakeholders.

Mechanical Engineer – Buildings, Infrastructure & Sustainability Groups

- Developed energy models to reduce building energy loads and inform design.
- Led design and analysis of mechanical scope on critical infrastructure projects globally.

BURNS & MCDONNELL - New York, NY, USA & Houston, TX, USA (May 2015 - August 2019)

Assistant Mechanical Engineer – Aviation & Federal Group

- Collaborated on front end planning through detailed design for mechanical systems on the $4B Delta LaGuardia Airport Terminal Reconfiguration Project, coordinating piping, HVAC and fueling scope across a large multidisciplinary infrastructure team.
- Automated mechanical design models, improving workflow efficiency for the project team across disciplines.

Assistant Chemical Engineer – Oil, Gas & Chemicals Group

- Facilitated front end planning through detailed design of oil and gas processing plant projects.
- Created automation processes for P&ID models to include data in heat and material balances to generate equipment, line list and pipe sizing calculations resulting in improved workflow efficiency across groups.

EDUCATION

CARNEGIE MELLON UNIVERSITY - Pittsburgh, PA, USA (December 2026, expected)

Master of Science, Chemical Engineering

Thesis: Context Driven Process Synthesis for Waste Valorization, advised by Ignacio Grossmann

TEXAS TECH UNIVERSITY - Lubbock, TX, USA (May 2015)

Bachelor of Science, Chemical Engineering (Minor: Bioengineering, Chemistry)

SKILLS

Software: Aspen (HYSYS, PLUS, etc.), IES, COMSOL, Smart Plant P&ID, Revit, UniSim, Excel, LLM Agents (Codex, Claude Code)
Programming: Python (Pyomo, IDAES, causal-learn, scikit-learn, JAX, etc.), GAMS, Matlab.

CONFERENCE PRESENTATIONS & PUBLICATIONS

- UKPACT. *eThekwini Regional Hydrogen Strategy.* eThekwini Municipality Energy Office, 2023. Link.
- Dempsey, K. & Gillespie, J. *" Can Synthetic Fuels Help Reduce Emissions in Australia's Rail Sector While Paving the Way to a Carbon Free Future?".* Presentation at *AUSRail,* 6 Dec. 2022, Brisbane, QLD. Conference Paper. Link.
- Department of Climate Change, Energy, the Environment and Water, and Australian Hydrogen Council. *Water for Hydrogen: Technical Paper.* Reference: 286739-00-RPT-001. 14 Nov. 2022, Link.
- Dempsey, K. *Cleaner, Alternative Fuels for Transport.* Presentation at *Our Pathway Through Net Zero: Reduce, Restore, Remove: A Call to Action,* 30 Aug. 2022, Melbourne, VIC. Video. Link.
- World Bank. *Opportunities and Barriers for the Deployment of Green Hydrogen in Chile's Markets: Small and Medium Grids: Task 3 - Isolated Grid Stud*y. Washington, D.C.: World Bank Group, 01 Aug. 2022. Link.